ICP 04. Thresholds and Delivery Timing

Companion reading handout for simultaneous listening.
Audio course on intrahepatic cholestasis of pregnancy. Episode four. Thresholds and delivery timing without memorization.

Source Base

This episode is based on the Israeli position paper on intrahepatic cholestasis of pregnancy, Ovadia two thousand nineteen, and the SMFM consult series on ICP.

The goal of this episode is to make the dry numbers memorable.

The numbers in ICP are ten, forty, and one hundred.

Ten names it. Forty changes the tempo. One hundred changes the risk conversation.

The Three Thresholds

Let us take them one by one.

Ten micromoles per liter is the diagnostic threshold most often used. If a pregnant patient has characteristic pruritus and total bile acids above ten, ICP is supported. This does not mean that a patient with bile acids of nine and classic symptoms is impossible. It means the diagnosis is not yet biochemically confirmed. If symptoms persist, repeat testing.

The number ten belongs to diagnosis. It does not by itself mean delivery. A patient at thirty-one weeks with bile acids of fourteen does not need immediate induction. She needs treatment for symptoms, repeat labs, risk counseling, and a plan.

Forty

Now forty.

Forty micromoles per liter is a management boundary. The Israeli position paper classifies mild disease up to forty and severe disease above forty. Ovadia and earlier cohort data show that higher bile-acid levels are associated with more adverse perinatal outcomes, including spontaneous preterm birth and meconium.

Forty is not the same as one hundred. But forty tells you that the pregnancy has moved out of the lowest-risk biochemical group. It changes the tempo of follow-up and delivery planning.

One Hundred

Now one hundred.

One hundred micromoles per liter is the high-risk stillbirth threshold. In Ovadia's individual participant data meta-analysis, stillbirth prevalence in singleton pregnancies was about zero point one three percent below forty, zero point two eight percent from forty to ninety-nine, and three point four four percent at one hundred or higher.

That is why one hundred is not just another round number. It is the point where the fetal-risk conversation changes.

There is one evidence caveat residents should understand. These are observational data from pregnancies managed in real clinical systems. Many patients were delivered early, especially when clinicians were worried. That active management may have helped keep stillbirth rates low in the lower bile-acid groups. So the correct interpretation is not, below one hundred is risk-free. The correct interpretation is, the clearest stillbirth signal is at one hundred or higher, and lower groups still require repeat testing and planned management.

Peak Value

Now let us talk about peak value.

Delivery timing is based on the highest bile-acid level measured, not simply the most recent value after treatment. This is explicit in the Israeli position paper, and it is consistent with Ovadia's conclusion that women should be managed according to maximum bile-acid concentration.

Why peak?

Because a later lower value may reflect treatment, timing of sampling, or natural fluctuation. It does not prove that the fetus was never exposed to the higher risk category. Peak bile acids are the pregnancy's demonstrated maximum exposure.

Delivery Timing

Now we move to delivery timing.

Israeli Guidance

Start with the local Israeli recommendation, because local practice matters.

If peak bile acids are below forty micromoles per liter, the recommended delivery window is thirty-seven plus zero to thirty-eight plus six.

If peak bile acids are forty to ninety-nine, the recommended delivery window is thirty-six plus zero to thirty-seven plus zero.

If peak bile acids are above one hundred, the recommended delivery window is thirty-four plus zero to thirty-six plus zero.

The Israeli paper also says that delivery at thirty-four plus zero to thirty-six plus zero may be considered when elevated bile acids are accompanied by pruritus unresponsive to treatment, a history of fetal death at an earlier gestational age in the presence of cholestasis, or departmental discretion.

SMFM Guidance

Now compare SMFM.

SMFM recommends delivery between thirty-six plus zero and thirty-nine plus zero for ICP with total bile acids below one hundred. Within that group, values below forty fit the later end, and values forty to ninety-nine fit earlier delivery.

For total bile acids at or above one hundred, SMFM recommends offering delivery at thirty-six plus zero, because the risk of stillbirth increases substantially around this gestational age.

SMFM reserves delivery between thirty-four and thirty-six weeks for selected patients with bile acids at or above one hundred plus additional high-risk features. Those features include excruciating and unremitting pruritus not relieved by medication, a prior stillbirth before thirty-six weeks due to ICP with recurrent ICP in the current pregnancy, or preexisting or acute hepatic disease with worsening function.

So how should the resident reconcile this?

Do not pretend the guidelines are identical. They are not. The Israeli guidance is more aggressive for the above-one-hundred group, allowing thirty-four to thirty-six weeks. SMFM generally defaults to thirty-six plus zero unless additional high-risk features exist.

But they share the same skeleton.

Bile acids below forty are the lower-risk group.

Bile acids forty to ninety-nine move delivery earlier.

Bile acids one hundred or higher are the high-risk group.

The difference is how aggressively each system balances prematurity against stillbirth risk.

Clinical Scenarios

Now let us make this practical with scenarios.

Scenario one. Thirty-two weeks, severe itching, bile acids eighteen, mild ALT elevation.

This is ICP, but it is not a delivery diagnosis at thirty-two weeks. Start ursodiol, repeat bile acids and transaminases, counsel, follow in a high-risk framework, and plan delivery according to later peak values.

Scenario two. Thirty-five plus three, bile acids fifty-two.

This is the forty to ninety-nine group. Under Israeli guidance, delivery is planned thirty-six plus zero to thirty-seven plus zero. SMFM would also move toward the earlier part of the thirty-six to thirty-nine range. This is where corticosteroids may matter if delivery is expected before thirty-seven weeks and they have not already been given.

Scenario three. Thirty-five plus one, bile acids one hundred and twenty.

This is the high-risk group. Under Israeli guidance, delivery from thirty-four to thirty-six weeks is recommended. Under SMFM, offer delivery at thirty-six plus zero, but consider thirty-four to thirty-six if there is unremitting pruritus, prior early ICP stillbirth, or worsening hepatic disease. In real practice, this patient belongs in MFM-level discussion, with explicit counseling about prematurity and fetal risk.

Scenario four. Thirty-seven plus two, classic pruritus, no rash, prior ICP, ALT elevated, bile acids pending.

Here, diagnostic certainty is incomplete, but gestational age is early term. SMFM allows shared decision-making in early-term suspected ICP when bile acids are pending, especially if transaminases are elevated or there is prior ICP. The Israeli paper says term delivery may be considered in high clinical suspicion without bile-acid testing. The key is shared decision-making, not pretending certainty.

Scenario five. Thirty-five weeks, itching, but bile acids normal and no repeat confirmation.

SMFM recommends against preterm delivery before thirty-seven weeks for clinical ICP without laboratory confirmation of elevated bile acids. That is an important safety rule. Treat symptoms if needed, repeat testing, and keep looking, but do not create iatrogenic prematurity without biochemical evidence.

The Emotional Side of Thresholds

Now let us talk about the emotional side of thresholds.

Patients and clinicians often want one number that says induce now. ICP does not work that way. Bile acids create a risk category. Gestational age determines what you can do with that risk. A bile acid of one hundred and five at thirty-four weeks is not the same operational problem as one hundred and five at thirty-seven weeks.

The decision is always a balance.

On one side is fetal exposure to high bile acids and the possibility of sudden fetal death.

On the other side is prematurity, respiratory morbidity, neonatal intensive care, and long-term effects of early birth.

Delivery timing is where those two risks are forced into the same room.

Summary

Now final synthesis.

First, ten diagnoses. In a symptomatic patient, bile acids above ten micromoles per liter support ICP.

Second, forty changes the tempo. It marks a higher-risk management group and moves delivery earlier.

Third, one hundred changes the danger category. It is the key stillbirth-risk threshold from Ovadia two thousand nineteen.

Fourth, below one hundred is not the same as no risk. It means lower risk under active clinical management, with repeat testing and planned delivery.

Fifth, use peak bile acids. Do not let a later treated value erase the highest exposure category.

Sixth, local Israeli guidance and SMFM differ in exact timing above one hundred, but they agree on the central biology of risk stratification.

Ten names it. Forty moves it earlier. One hundred changes the risk conversation.

End of episode four.